PAPER ANALYSIS

Counterexamples to Five Conjectured Upper Bounds

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Title Page (cont. 1/29)

ORIGINAL (ENGLISH)

Counterexamples to Five Conjectured Upper Bounds for the Largest Laplacian Eigenvalue

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Title Page — Role in Overall Structure

This is the **opening frame** of the paper. It identifies the problem domain (spectral graph theory), the contribution type (counterexamples), and the author. The two tables previewed here serve as a roadmap: Table 1 summarizes found counterexamples; Table 2 catalogs the 33 bounds still open.

- **Laplacian eigenvalue**: largest eigenvalue λ_1 of a graph's Laplacian matrix $L = D - A$
- **BHS bounds**: a family of conjectured upper bounds for λ_1, referenced throughout
- **Gap**: difference between bound value and actual λ_1 — negative gap = counterexample

> Think of BHS bounds as speed limit signs for λ_1. This paper finds roads where cars legally exceed those limits — disproving the signs, not breaking laws.

> Providing explicit counterexamples is the sharpest way to falsify conjectures — no probabilistic argument, just a single graph that violates the bound.

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Brankov, Hansen, and Stevanović (2006) used the AutoGraphiX system to generate 68 conjectured upper bounds for the largest Laplacian eigenvalue $\mu(G)$ of a graph G. Ghebleh et al. (2024) disproved 30 of these, leaving 38 open. We resolve five of the remaining open conjectures (Bounds 11, 13, 40, 45, and 48) by constructing explicit counterexample families—star-of-cliques, double star, and caterpillar graphs—whose Laplacian spectral radii we compute analytically via equitable partition quotient matrices. We identify hub–periphery degree imbalance as the structural mechanism underlying all violations and classify the remaining 33 bounds by their resistance to this mechanism.

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What This Section Does

This is the paper's **abstract/contribution summary**. It situates the work within a known open-problem lineage and states precisely what is proved. As the opening section, it previews every major concept developed in later sections.

- $\mu(G)$: largest (spectral radius) eigenvalue of the **Laplacian matrix** of graph G
- **Equitable partition quotient matrix**: a reduced matrix that shares eigenvalues with the full Laplacian for symmetric graphs
- **Hub-periphery degree imbalance**: structural property where one high-degree node connects to many low-degree leaves

> Think of the 68 conjectures as a leaderboard of unverified world records. Ghebleh et al. knocked out 30 records as fake; this paper challenges 5 more by producing actual athletes (counterexample graph families) who beat the claimed limits.

Concept Relationships: Since this is the first annotated section, it maps the whole paper's arc: the conjecture catalog (prior work) → counterexample construction (Method) → analytic eigenvalue computation via quotient matrices (Results) → structural classification of remaining bounds (Discussion).

> Five long-open bounds fall not from abstract algebra alone, but because a single geometric motif — a dense hub radiating to sparse periphery — consistently inflates $\mu(G)$ beyond what the conjectured formulas allow.

Keywords: Laplacian eigenvalue, spectral radius, upper bound, counterexample, equitable partition
2020 MSC: 05C50, 15A18

1. Introduction

Let $G = (V, E)$ be a simple connected graph on $n = |V|$ vertices and $m = |E|$ edges. The *Laplacian matrix* of G is $L(G) = D(G) - A(G)$, where $A(G)$ is the adjacency matrix and $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$ is the diagonal matrix of vertex degrees. Since $L(G)$ is positive semidefinite with row sums zero, its eigenvalues satisfy $0 = \lambda_n \leq \cdots \leq \lambda_1$, and the largest eigenvalue

$$\mu(G) := \lambda_{\max}(L(G)) \tag{1}$$

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Section 1 — Introduction: Graph & Laplacian Setup

Variable / Symbol Explanation — for **Equation (1)**:

- $G = (V, E)$: simple connected graph; $n = |V|$ vertices, $m = |E|$ edges
- $A(G)$: adjacency matrix — entry 1 if edge exists, 0 otherwise
- $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$: diagonal degree matrix
- $L(G) = D(G) - A(G)$: Laplacian matrix (positive semidefinite, row-sums zero)
- $0 = \lambda_n \leq \cdots \leq \lambda_1$: eigenvalues in non-decreasing order

$$\mu(G) := \lambda_{\max}(L(G))$$

Think of $L(G)$ as a "tension matrix": it measures how unevenly connections pull on each vertex. $\mu(G)$, the spectral radius, is the worst-case tension — the paper's central object to bound.

Relationship to Previous Sections: The keywords on the title page named **spectral radius** and **upper bound** as targets; this paragraph defines exactly what *spectral radius* means ($\mu(G)$), grounding those keywords in formal notation before the paper's bounds are stated.

The spectral radius $\mu(G)$ encodes the graph's global connectivity extremes, making its upper bound a powerful structural invariant — that is why bounding it (and finding counterexamples to known bounds) is the paper's core contribution.

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is called the *Laplacian spectral radius*. Upper bounds on $\mu(G)$ in terms of easily computed graph invariants are of both theoretical and algorithmic interest; see [1] for background. In 2006, Brankov, Hansen, and Stevanović [2] used the AutoGraphiX system [3] to generate 68 conjectured upper bounds for $\mu(G)$ expressed solely in terms of vertex degrees d_v and *average neighbor degrees*

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u. \tag{2}$$

The 68 conjectures fall into two families. *Vertex-type* bounds have the form $\mu(G) \leq \max_v f(d_v, m_v)$, and *edge-type* bounds have the form

$$\mu(G) \leq \max_{ij \in E} g(d_i, d_j, m_i, m_j).$$

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Laplacian Spectral Radius & Conjectured Bounds

Variable/Symbol Explanation — Equation (2):

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u$$

- $\mu(G)$: **Laplacian spectral radius** — the largest eigenvalue of $L(G)$
- d_v: degree of vertex v (number of its neighbors)
- m_v: **average neighbor degree** of v
- $u \sim v$: sum over all neighbors u of v

Think of d_v as how popular a person is, and m_v as the average popularity of their friends. A vertex with few friends who are all well-connected has a high m_v despite a low d_v.

Concept Relationships: The previous section introduced $L(G) = D - A$ and defined $\mu(G)$ as its largest eigenvalue. This section now asks: can we *bound* $\mu(G)$ without full eigenvalue computation, using only local quantities d_v and m_v?

The **AutoGraphiX** system (2006) computationally searched over graphs to conjecture 68 such bounds, split into two families:

- **Vertex-type:** $\mu(G) \leq \max_v f(d_v, m_v)$ — optimized over vertices
- **Edge-type:** $\mu(G) \leq \max_{ij \in E} g(d_i, d_j, m_i, m_j)$ — optimized over edges

Bounding a global spectral quantity by local degree statistics is powerful: it makes $\mu(G)$ tractable for large graphs where full eigendecomposition is costly.

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None had been proved or disproved for nearly two decades.

Ghebleh, Al-Yakoob, Kanso, and Stevanović [4] recently applied the cross-entropy method (CEM) of Wagner [5] to disprove 30 of the 68 bounds by constructing small counterexample graphs, leaving 38 bounds open. Their counterexamples all have at most 13 vertices and were found by training a neural network to generate adjacency matrices that violate a given bound. Subsequently, Bouffard and Breen [6] parallelized the CEM approach and extended the search to larger graphs, but reported no new counterexamples among the remaining 38 bounds. Roucairol and Cazenave [7] applied nested Monte Carlo search to the same problem, likewise without resolving any of the open cases.

In the present paper we disprove five of the 38 remaining open bounds.

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Research Progress: Narrowing 68 Open Bounds

This paragraph is the **literature review** portion of the Introduction, tracing the stepwise reduction of open conjectures before the present paper's contribution.

Prior Work Timeline:

- ~20 years : 68 bounds untouched — no proof or disproof
- **Ghebleh et al.**: CEM + neural network → disproved **30**, leaving **38 open**
- **Bouard & Breen**: parallelized CEM, searched larger graphs → no new counterexamples
- **Roucairol & Cazenave**: nested Monte Carlo search → still no new counterexamples
- **This paper**: disproves **5** of the 38 remaining bounds

> Think of it like a decades-old cold case file: Ghebleh's team cracked 30 cases with AI forensics, but the last 38 stumped two follow-up investigations. This paper reopens the file and resolves 5 more — not by brute force, but presumably by a different technique (revealed later).

Connection to previous sections: Earlier in the Introduction, the 68 bounds were introduced as specific spectral graph theory conjectures. This paragraph now maps the exact state of knowledge going into the present work — establishing that the 38 remaining bounds are the hard cases that resisted both ML-based search and Monte Carlo methods.

> The key tension is that Ghebleh's neural-network approach hit a wall at 38 bounds — suggesting the remaining counterexamples (if they exist) either

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Theorem 1.1 (Main result). *BHS Bounds 11, 13, 40, 45, and 48 from [2] are false. Specifically:*

(a) *Bound 11 is violated by* $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 4$ *and* $t \geq 2m - 1$ (Theorem 4.2).

(b) *Bound 13 is violated by* $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and sufficiently large* t (Theorem 4.3).

(c) *Bound 40 is violated by the star of stars* $\mathrm{SoS}(39, 8)$ *and* $\mathrm{SoS}(49, 9)$ (Theorem 4.7).

(d) *Bound 45 is violated by* $\mathrm{DS}(k, k)$ *for all* $k \geq 6$ (Theorem 4.5).

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Theorem 1.1 — Main Result

This is the paper's central claim: five specific **BHS Bounds** from the reference work [2] are disproven by explicit counterexamples. Each sub-item pairs a bound number with a graph family that violates it.

- **SoC(K_m, t)**: Star of Cliques — a clique K_m used as the "center" structure, with parameter t
- **SoS(n, k)**: Star of Stars — nested star graph with specific size parameters
- **DS(k, k)**: a symmetric double-star (or similar) graph on parameter k
- **P_3**: the simple path on 3 vertices — the simplest counterexample (Bound 48)

> Think of BHS Bounds as a rulebook claiming "no graph can exceed this score." This theorem is like finding five actual games where the score was exceeded — the rulebook is wrong, and the counterexamples are the scorecards.

Relation to prior sections: The Abstract and Introduction (sections 1–5/29) established the problem setting and defined BHS Bounds as conjectured upper limits in some graph parameter. This theorem is the payoff — it directly refutes those conjectures with parameterized families, each proved in a dedicated later theorem (4.2, 4.3, 4.5, 4.7, 4.8).

> The use of infinitely many counterexamples (e.g., all $m \geq 4, t \geq 2m - 1$) means the bounds fail structurally, not just in isolated edge cases.

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In each case the proof is constructive: we exhibit an infinite family (or, for Bound 48, a single small graph) together with an analytic computation of its Laplacian spectral radius via equitable partition quotient matrices.

Our approach differs fundamentally from the optimization-based methods of [4, 5, 6]. Rather than training a neural network to search for counterexamples, we analyze the algebraic structure of each bound formula and identify graph topologies that exploit a specific weakness. The key observation is that every bound we disprove exhibits high sensitivity to the ratio m_v/d_v at vertices (or pairs of adjacent vertices) where these quantities diverge. Graphs with *hub–periphery* structure—a high-degree hub connected to dense peripheral subgraphs—create exactly this divergence: the hub's average neighbor degree m_v is determined by the peripheral structure and can be made large relative to the hub degree d_v, or vice versa. Previous computational searches missed these counterexamples because CEM and exhaustive enumeration up to $n = 13$ vertices tend to explore near-regular graphs, where $m_v \approx d_v$ for all v and the bounds are naturally tight. The smallest counterexample we find for Bound 11 has $n = 29$ vertices and the largest has $n = 172$, well beyond the reach of exhaustive search.

The paper is organized as follows. Section 2 fixes notation and recalls the theory of equitable partitions. It also states the five BHS bounds that we disprove. Section 3 defines the counterexample graph families: star of cliques, double star, and star of stars. Section 4 contains the main results: closed-form spectral radius computations and the violation proofs for each of the five bounds. Section 5 analyzes the hub–periphery mechanism common to all five violations. Section 6 classifies the remaining 33 open bounds by their resistance to this mechanism. Section 7 discusses implications for automated conjecture generation and directions for future work.

EXPLANATION (ENGLISH)

The proofs are **constructive**: explicit infinite graph families (or a single small graph for Bound 48) are exhibited, and their **Laplacian spectral radius** is computed analytically via **equitable partition quotient matrices**.

This approach contrasts with optimization-based methods [4,5,6] that train neural networks to search for counterexamples. Here, the algebraic structure of each bound formula is analyzed directly to find topologies that exploit a specific weakness.

Key observation: every disproved bound shows high sensitivity to the ratio m_v/d_v at vertices (or adjacent vertex pairs) where these quantities diverge. **Hub-periphery graphs** — a high-degree hub connected to a dense periphery — are the primary exploiting topology.

- m_v: average degree of neighbors of vertex v
- d_v: degree of vertex v
- m_v/d_v: ratio that diverges in hub-periphery configurations

> Think of each bound formula as a bridge designed for uniform traffic. Hub-periphery graphs are like sending one enormous truck while the formula assumed only bicycles — the structural asymmetry breaks the design assumption.

Relation to previous sections: The prior sections introduced the bounds and stated they would be disproved. This section reveals *how*: not by black-box search, but by algebraic diagnosis — finding the ratio m_v/d_v as the universal fault line across all failing bounds.

> The core insight is that counterexamples are not found by luck or search, but derived from the formula's own algebraic sensitivity — making each disproof structurally explanatory, not just existential.

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Title Page (cont.) — Why Prior Searches Failed & Paper Roadmap

Key variables in the divergence argument:

- d_v: degree of vertex v (hub) — number of its direct neighbors
- m_v: average degree of v's neighbors — determined by the peripheral structure
- **Star-of-stars**: the graph family exploiting the gap $m_v \gg d_v$ (or vice versa)

The core mechanism: in a hub-and-spoke topology, m_v and d_v decouple. The hub has moderate degree d_v, but its neighbors (spokes) each connect to many leaves, inflating m_v independently. This decoupling is exactly what the BHS bounds fail to anticipate.

> Think of a regional airport (hub) connected to small local airports (spokes), each of which serves many tiny airstrips. The hub's own connection count (d_v) is modest, but the average "busyness" of its neighbors (m_v) can be enormous — or tiny — depending on how many airstrips each spoke serves. Near-regular graphs are like a uniform grid of airports: every node is equally busy, so the gap never appears.

Why CEM and exhaustive search missed this: Both methods explore near-regular graphs (where $m_v \approx d_v$ for all v), the regime where the BHS bounds are naturally tight. The counterexamples live at $n = 29$–172, far outside exhaustive-search range ($n \leq 13$).

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EXPLANATION (ENGLISH)

Relationship to previous sections: The introduction established that BHS bounds were computationally unchallenged for decades. This passage explains the *structural reason* they survived so long — not because they are correct, but because standard search tools are blind to the hub-peripheral divergence.

Paper roadmap:

- **§ 2**: Notation + equitable partition theory + the five BHS bounds to be disproved
- **§ 3**: Defines the counterexample graph families (star of …)

> The structural insight is that irregular graphs with a deliberate hub-periphery asymmetry form a blind spot for exhaustive enumeration, which is why a decades-old conjecture survived unchallenged until targeted construction.

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2. Preliminaries

2.1. Graph notation

Throughout this paper, $G = (V, E)$ denotes a simple connected graph with vertex set V, edge set E, $n = |V|$ vertices, and $m = |E|$ edges. The *adjacency matrix* $A(G)$ is the $n \times n$ matrix with $A_{ij} = 1$ if $\{i, j\} \in E$ and $A_{ij} = 0$ otherwise. The *degree matrix* is $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$, where $d_v = \sum_u A_{vu}$ is the degree of vertex v. The *Laplacian matrix* $L(G) =$

This section serves two roles: a **roadmap** of the paper's core contributions (Sections 4–7) and the foundational **graph notation** used throughout.

Section Roadmap (Sections 4–7)

- **§ 4**: Closed-form spectral radius computations + violation proofs for 5 bounds
- **§ 5**: Hub-periphery mechanism — the shared structure behind all 5 violations
- **§ 6**: Classification of 33 remaining open bounds by resistance to that mechanism
- **§ 7**: Implications for automated conjecture generation

Graph Notation

- $G = (V, E)$: simple connected graph with vertex set V, edge set E
- $n = |V|$: number of vertices; $m = |E|$: number of edges
- $A(G)$: $n \times n$ adjacency matrix, $A_{ij} = 1$ if $\{i, j\} \in E$, else 0
- $D(G) = \mathrm{diag}(d_1, \ldots, d_n)$: degree matrix, $d_v = \sum_u A_{vu}$

$$A_{ij} = \begin{cases} 1 & \{i, j\} \in E \\ 0 & \text{otherwise} \end{cases}$$

Think of $A(G)$ as a friendship table: rows and columns are people, and a 1 means they're connected. $D(G)$ is simply a diagonal summary of how many friends each person has.

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Relationship to previous sections: Section 1 introduced the three graph families (cliques, double star, star of stars) used as counterexamples. This Preliminaries section now establishes the shared mathematical language — $A(G), D(G), n, m$ — that Sections 4–6 will use to compute spectral radii and verify bound violations on those families.

> These two matrices, A and D, are the building blocks for nearly every spectral graph quantity (Laplacian, normalized Laplacian, signless Laplacian) that the 38 bounds involve.

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$D(G) - A(G)$ is positive semidefinite with smallest eigenvalue 0. We write $\mu(G) = \lambda_{\max}(L(G))$ for the *Laplacian spectral radius*.

For each vertex $v \in V$ with $d_v \geq 1$, the *average neighbor degree* is

$$m_v = \frac{1}{d_v} \sum_{u \sim v} d_u,$$

where the sum is over all neighbors u of v. Note that $m_v \geq 1$ for every vertex in a connected graph, and $m_v = d_v$ for every vertex if and only if G is regular.

2.2. Equitable partitions

Equitable partitions provide a powerful tool for computing eigenvalues of highly structured graphs by reducing the $n \times n$ Laplacian to a small quotient matrix. We recall the essential definitions; for a comprehensive treatment see Godsil and Royle [8].

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§ 2.1 cont. — Laplacian Matrix & Average Neighbor Degree; § 2.2 — Equitable Partitions

Variable/Symbol Explanation

- d_v: degree of vertex v (number of edges incident to v)
- $D(G)$: diagonal degree matrix; $A(G)$: adjacency matrix (from § 2.1)
- $\mu(G) = \lambda_{\max}(L(G))$: **Laplacian spectral radius** — largest eigenvalue of L
- m_v: average neighbor degree of v

$$L(G) = D(G) - A(G), \quad m_v = \frac{1}{d_v} \sum_{u \sim v} d_u$$

$L(G)$ is positive semidefinite : all eigenvalues ≥ 0, with the smallest always equal to 0 (constant vector is in the null space). The quantity $\mu(G)$ captures how "spread out" the graph's connectivity is. Note $m_v = d_v$ for all v iff G is regular (every neighbor has the same degree as v).

Think of m_v as a vertex asking its neighbors: "What's your average popularity?" In a regular graph, everyone has equal popularity, so the answer always matches your own degree.

ORIGINAL (ENGLISH)

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Connection to Previous Sections: $D(G)$ and $A(G)$ were introduced in § 2.1; the Laplacian $L = D - A$ combines both into a single matrix encoding the full spectral structure of G. The spectral radius $\mu(G)$ is the key quantity this paper aims to bound.

Equitable Partitions (§ 2.2): partition V into cells so each vertex in cell C_i has the **same number of neighbors** in every cell C_j. This regularity lets the $n \times n$ Laplacian be compressed to a small quotient matrix whose eigenvalues are also eigenvalues of $L(G)$.

> Like summarizing a large symmetric org-chart by department — once each department behaves uniformly toward others, you only need the department-level summary to recover key spectral information.

> Equitable partitions are the bridge that makes exact eigenvalue computation tractable for structured graphs — reducing an $n \times n$ problem to a tiny matrix.

Definition 2.1 (Equitable partition). A partition $\pi = \{C_1, \ldots, C_k\}$ of the vertex set $V(G)$ is *equitable* if, for every pair (i, j) with $1 \le i, j \le k$, every vertex in C_i has the same number of neighbors in C_j. We denote this common value by b_{ij}.

Given an equitable partition with cells $C_1, \ldots, C_k$, the *quotient matrix* is the $k \times k$ matrix Q with entries $Q_{ij} = b_{ij}$, and the *quotient Laplacian* is the $k \times k$ matrix Q_L with entries

$$(Q_L)_{ij} = \begin{cases} d_{C_i} & \text{if } i = j, \\ -b_{ij} & \text{if } i \neq j, \end{cases}$$

where $d_{C_i} = \sum_{j=1}^{k} b_{ij}$ is the common degree of every vertex in cell C_i.

Theorem 2.2 ([8, Theorem 9.3.3]). *Let π be an equitable partition of G with quotient Laplacian Q_L. Then every eigenvalue of Q_L is an eigenvalue of $L(G)$. In particular,*

$$\mu(G) \ge \lambda_{\max}(Q_L),$$

Definition 2.1 & Theorem 2.2 — Equitable Partition & Quotient Laplacian

Equitable partition $\pi = \{C_1, \ldots, C_k\}$ groups vertices so every vertex in cell C_i has exactly the same number of neighbors in C_j, captured by the constant b_{ij}.

- π: partition of vertex set $V(G)$ into k cells
- b_{ij}: common neighbor-count from any vertex in C_i into C_j
- Q: $k \times k$ **quotient matrix**, $Q_{ij} = b_{ij}$
- $d_{C_i} = \sum_{j=1}^{k} b_{ij}$: uniform degree of every vertex in C_i
- $\mu(G)$: algebraic connectivity (second-smallest eigenvalue of $L(G)$)

> $$(Q_L)_{ij} = \begin{cases} d_{C_i} & i = j \\ -b_{ij} & i \neq j \end{cases}$$

This mirrors the full Laplacian $L(G)$ (introduced in prior sections) but collapsed to $k \times k$ — same diagonal-degree / off-diagonal-negative structure, just over cells instead of individual vertices.

> Think of Q_L as a *zoomed-out photo* of $L(G)$: cells replace individual pixels, but the Laplacian pattern (diagonal = degree, off-diagonal = −edge weight) is preserved exactly.

Theorem 2.2 states every eigenvalue of Q_L is also an eigenvalue of $L(G)$, yielding the lower bound:

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Definition 2.1 (Equitable partition). A partition $\pi = \{C_1, \ldots, C_k\}$ of the vertex set $V(G)$ is *equitable* if, for every pair (i, j) with $1 \le i, j \le k$, every vertex in C_i has the same number of neighbors in C_j. We denote this common value by b_{ij}.

Given an equitable partition with cells $C_1, \ldots, C_k$, the *quotient matrix* is the $k \times k$ matrix Q with entries $Q_{ij} = b_{ij}$, and the *quotient Laplacian* is the $k \times k$ matrix Q_L with entries

$$(Q_L)_{ij} = \begin{cases} d_{C_i} & \text{if } i = j, \\ -b_{ij} & \text{if } i \neq j, \end{cases}$$

where $d_{C_i} = \sum_{j=1}^{k} b_{ij}$ is the common degree of every vertex in cell C_i.

Theorem 2.2 ([8, Theorem 9.3.3]). *Let π be an equitable partition of G with quotient Laplacian Q_L. Then every eigenvalue of Q_L is an eigenvalue of $L(G)$. In particular,*

$$\mu(G) \ge \lambda_{\max}(Q_L),$$

EXPLANATION (ENGLISH)

$$\mu(G) \ge \lambda_{\max}(Q_L)$$

By constructing a small equitable partition whose Q_L has a large maximum eigenvalue, one can certify a strong lower bound on $\mu(G)$ without computing the full spectrum of $L(G)$.

ORIGINAL (ENGLISH)

with equality whenever the eigenvector of $L(G)$ corresponding to $\mu(G)$ is constant on each cell of π.

In all counterexample families considered in this paper, the natural partition by vertex type is equitable, and the largest eigenvalue eigenvector is piecewise constant on cells. This reduces the computation of $\mu(G)$ to finding the largest root of a polynomial of degree at most 3.

EXPLANATION (ENGLISH)

§ 2.2 (cont.) Equitable Partitions & § 2.3 BHS Bounds

This section has two roles: (1) closing the computational shortcut from § 2.2, and (2) stating the five BHS **conjectured upper bounds** on $\mu(G)$ that the paper disproves.

Equations

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^3}{d_v^2} \quad \text{(BHS Bound 11)}$$

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3} \quad \text{(BHS Bound 13)}$$

- $\mu(G)$: largest eigenvalue of the Laplacian $L(G)$ (target quantity from § 2.1)
- d_v: degree of vertex v (edge count)
- m_v: edge count among the neighbors of v (local density)
- V: vertex set of G

Both bounds are vertex-type bounds — they depend only on local neighborhood statistics at each vertex.

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2.3. The BHS bounds

We now state the five conjectured bounds from [2] that we disprove. All five are listed as open in [4, Table 1]. Bounds 11 and 13 are vertex-type bounds; Bounds 40, 45, and 48 are edge-type bounds.

Conjecture 2.3 (BHS Bound 11 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{v \in V} \frac{2\, m_v^3}{d_v^2}.$$

Conjecture 2.4 (BHS Bound 13 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{v \in V} \frac{2\, m_v^4}{d_v^3}.$$

EXPLANATION (ENGLISH)

Think of $\mu(G)$ as the "loudest resonance" of a network. Bounds 11 and 13 claim this resonance can't exceed a threshold computed from each vertex's local density ratio — like saying a speaker's peak volume is capped by its wiring specs alone. The paper's counterexamples show some graphs "exceed spec."

Connection to previous sections: § 2.2 introduced **equitable partitions** and showed $\mu(G)$ equals the largest root of a low-degree polynomial when the Fiedler eigenvector is piecewise constant on cells. The counterexample families used here exploit exactly that structure — their natural vertex-type partition is equitable, reducing eigenvalue computation to a degree-$\leq$3 polynomial root search. This makes it feasible to verify violations of Bounds 11/13/40/45/48 analytically.

Bounds 11 and 13 follow a pattern $\frac{2m_v^{k+1}}{d_v^k}$, suggesting a family of conjectured inequalities — disproving two of them hints the entire pattern may be flawed.

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Conjecture 2.5 (BHS Bound 40 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right).$$

Conjecture 2.6 (BHS Bound 45 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{(d_i-d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right).$$

Conjecture 2.7 (BHS Bound 48 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}.$$

EXPLANATION (ENGLISH)

Conjectures 2.5–2.7: BHS Upper Bounds on Spectral Radius

These three conjectures from [2] propose progressively simplified upper bounds on $\mu(G)$, all sharing the edge-maximum form introduced in earlier bounds.

Reconstructed equations:

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i - 1)^2 + 2(m_j - 1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right) \quad \text{(Bound 40}$$

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right) \quad \text{(Bound 45)}$$

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}} \quad \text{(Bound 48)}$$

- $\mu(G)$: spectral radius (largest adjacency eigenvalue)
- d_i, d_j: degrees of endpoints of edge (i, j)
- m_i, m_j: average neighbor degrees of vertices i, j

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Conjecture 2.5 (BHS Bound 40 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right).$$

Conjecture 2.6 (BHS Bound 45 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{(d_i-d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right).$$

Conjecture 2.7 (BHS Bound 48 [2]). *For every connected graph G,*

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}.$$

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Like three weather forecasts of increasing coarseness — Bound 40 uses fine-grained local data, Bound 48 drops to degree-only inputs, trading precision for simplicity.

Relation to previous sections: Like Conjectures 2.1–2.4, these bounds maximize over edges $(i, j) \in E$ and mix d_i, m_i parameters. The progression 40→45→48 reduces radicand complexity, converging toward degree-only expressions.

Each simplification tests whether cruder local data still controls the global spectral radius — a key question for practical, computable bounds.

ORIGINAL (ENGLISH)

Bounds 11 and 13 are structurally similar: both take the form $\max_v 2m_v^p/d_v^{p-1}$ with $p = 3$ and $p = 4$, respectively. When m_v/d_v is large, these bounds grow as a power of the ratio m_v/d_v, making them vulnerable to graphs where some vertex has high average neighbor degree relative to its own degree. Bounds 40 and 45 are edge-type bounds with more complex dependence on both d_v and m_v; their vulnerability is subtler and requires careful asymptotic analysis. Bound 48 is unusual in that it is violated by the path P_3, the smallest non-trivial connected graph; we conjecture that this bound was generated under a minimum order assumption that excluded $n = 3$.

EXPLANATION (ENGLISH)

Structural Analysis of Bounds & Counterexample Setup

Variable/Symbol Explanation

- m_v: average degree of neighbors of vertex v
- d_v: degree of vertex v itself
- p: exponent parameter (3 or 4 for Bounds 11/13)
- P_3: path graph on 3 vertices — the smallest nontrivial connected graph

Bounds 11 and 13 share the structural form:

$$\max_v \frac{m_v^{p-1}}{d_v^{p-1}}$$

with $p = 3$ and $p = 4$ respectively. When the ratio m_v/d_v is large (a hub vertex surrounded by high-degree neighbors), these bounds grow as a **power of the ratio**, making them fragile on such graphs.

> Think of m_v/d_v as a "neighborhood prestige ratio" — a low-degree vertex connected only to high-degree hubs inflates this ratio, like a small-town resident whose only friends are celebrities. Raising this ratio to a power amplifies the vulnerability rapidly.

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Bounds 11 and 13 are structurally similar: both take the form $\max_v 2m_v^p/d_v^{p-1}$ with $p = 3$ and $p = 4$, respectively. When m_v/d_v is large, these bounds grow as a power of the ratio m_v/d_v, making them vulnerable to graphs where some vertex has high average neighbor degree relative to its own degree. Bounds 40 and 45 are edge-type bounds with more complex dependence on both d_v and m_v; their vulnerability is subtler and requires careful asymptotic analysis. Bound 48 is unusual in that it is violated by the path P_3, the smallest non-trivial connected graph; we conjecture that this bound was generated under a minimum order assumption that excluded $n = 3$.

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Concept Relationships: Earlier sections catalogued these bounds and their claimed domains. Here, the analysis transitions from classification to structural weakness diagnosis — grouping bounds by their failure mode rather than their formula. Bounds 40 and 45 are **edge-type bounds** with joint dependence on both d_v and m_v, requiring asymptotic analysis to expose vulnerabilities (unlike the clean power-law argument for 11/13). Bound 48 is flagged as potentially ill-formed — violated by P_3 — suggesting it was derived under an implicit **minimum order assumption** ($n \geq 4$) that was never stated.

> A bound violated by the smallest possible graph is not a bound at all — identifying such cases clears the field before constructing systematic counterexamples in Section 3.

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Definition 3.1 (Star-of-Cliques). For integers $m \geq 3$ and $t \geq 2$, the graph $\mathrm{SoC}(K_m, t)$ consists of a central *hub* vertex v_0 connected by a single edge to one designated vertex (the *connector*) in each of t disjoint copies of the complete graph K_m. The total number of vertices is $n = 1 + tm$.

Definition 3.2 (Double Star). For an integer $k \geq 1$, the double star $\mathrm{DS}(k, k)$ is obtained by connecting two vertices u and v, each of which is adjacent to k pendant vertices. The total number of vertices is $n = 2k + 2$.

Definition 3.3 (Star-of-Stars). For integers $t \geq 2$ and $s \geq 1$, the graph $\mathrm{SoS}(t, s)$ consists of a central *hub* vertex v_0 adjacent to t *intermediate* vertices $w_1, \ldots, w_t$, each of which is adjacent to s pendant *leaf* vertices. The total number of vertices is $n = 1 + t + ts = 1 + t(1 + s)$.

EXPLANATION (ENGLISH)

Section 3 — Graph Family Definitions

This section formally defines three **graph families** used as test structures in the paper's main results. Each builds on a hierarchical hub-and-spoke pattern with increasing complexity.

Variable/Symbol Explanation

- m: clique size (number of vertices per complete subgraph K_m)
- t: number of clique/branch copies attached to the hub
- k: number of pendant (leaf) vertices per center in $DS(k, k)$
- s: number of leaves per intermediate vertex in $SoS(t, s)$
- v_0: central hub; w_i: intermediate vertices; n: total vertex count

> Think of all three graphs as airport networks. **SoC**: a hub connecting to dense city clusters (complete graphs). **DS**: two regional hubs, each serving local stops. **SoS**: a hub → regional airports → small airstrips — a two-level tree with leaves only at the bottom.

Structural Comparison

ORIGINAL (ENGLISH)

Definition 3.1 (Star-of-Cliques). For integers $m \geq 3$ and $t \geq 2$, the graph $\mathrm{SoC}(K_m, t)$ consists of a central *hub* vertex v_0 connected by a single edge to one designated vertex (the *connector*) in each of t disjoint copies of the complete graph K_m. The total number of vertices is $n = 1 + tm$.

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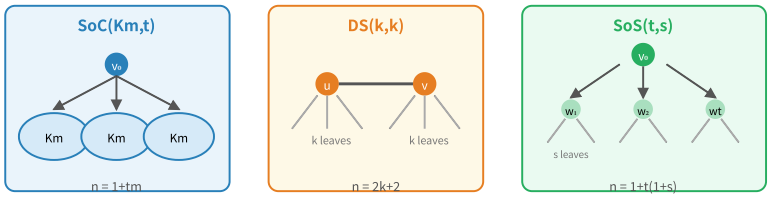
EXPLANATION (ENGLISH)



Relationship to Previous Sections: Earlier sections established combinatorial or graph-theoretic bounds in general terms. These three definitions provide the concrete graph families on which those bounds will be tested or proven tight. They form a complexity ladder: DS (depth 2, symmetric) → SoS (depth 3, uniform leaves) → SoC (depth 2, dense clique clusters).

> Each definition isolates a distinct structural feature — clique density, symmetric branching, or two-level hierarchy — allowing the paper's results to pinpoint exactly which graph property drives the behavior being studied.

Definition 3.4 (Caterpillar)**.** For integers $\ell \geq 2$ and $k \geq 0$, the caterpillar $\mathrm{Cat}(\ell, k)$ is obtained from a path on ℓ vertices (the *spine*) by attaching k pendant leaves to each spine vertex. The total number of vertices is $n = \ell(k+1)$.

4. Main results

Proposition 4.1. *For $m \geq 3$ and $t \geq 2$, the Laplacian spectral radius of $\mathrm{SoC}(K_m, t)$ is*

$$\mu\big(\mathrm{SoC}(K_m, t)\big) = \frac{(m+t+1) + \sqrt{(t-m)^2 + 2(t+m) - 3}}{2}. \tag{3}$$

Proof. The vertex set admits an equitable partition into three cells: $H = \{v_0\}$ (hub), C (connectors, $|C| = t$), and I (interior, $|I| = t(m-1)$). The quotient Laplacian is the 3×3 matrix

$$Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & m & -(m-1) \\ 0 & -1 & 1 \end{pmatrix}. \tag{4}$$

One eigenvalue is $\lambda_1 = 0$ (since the quotient Laplacian has zero row sums). The remaining two eigenvalues are roots of

$$\lambda^2 - (m+t+1)\lambda + (tm+1) = 0,$$

Definition 3.4 & Proposition 4.1: Caterpillar Graph and Laplacian Spectral Radius of SoC

Caterpillar Cat(ℓ, k): a path of ℓ spine vertices, each with k pendant leaves attached. Total vertices: $n = \ell(k+1)$.

Proposition 4.1 gives the Laplacian spectral radius of $\mathrm{SoC}(K_m, t)$:

$$\mu(\mathrm{SoC}(K_m, t)) = \frac{(m+t+1) + \sqrt{(t-m)^2 + 2(t+m) - 3}}{2} \tag{3}$$

- m: clique size of K_m (interior structure), $m \geq 3$
- t: number of connectors (copies attached), $t \geq 2$
- μ: largest eigenvalue of the Laplacian matrix

The proof partitions vertices into three **equitable partition** cells: hub $H = \{v_0\}$, connectors C ($|C| = t$), interior I ($|I| = t(m-1)$). The **quotient Laplacian** is:

$$Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & m & -(m-1) \\ 0 & -1 & 1 \end{pmatrix} \tag{4}$$

One eigenvalue is $\lambda_1 = 0$ (row sums are zero). The other two solve:

$$\lambda^2 - (m+t+1)\lambda + (tm+1) = 0$$

ORIGINAL (ENGLISH)

Definition 3.4 (Caterpillar)**.** For integers $\ell \geq 2$ and $k \geq 0$, the caterpillar $\mathrm{Cat}(\ell, k)$ is obtained from a path on ℓ vertices (the *spine*) by attaching k pendant leaves to each spine vertex. The total number of vertices is $n = \ell(k+1)$.

4. Main results

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Proof. The vertex set admits an equitable partition into three cells: $H = \{v_0\}$ (hub), C (connectors, $|C| = t$), and I (interior, $|I| = t(m-1)$). The quotient Laplacian is the 3×3 matrix

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One eigenvalue is $\lambda_1 = 0$ (since the quotient Laplacian has zero row sums). The remaining two eigenvalues are roots of

$$\lambda^2 - (m+t+1)\lambda + (tm+1) = 0,$$

EXPLANATION (ENGLISH)

Think of the equitable partition like a corporate org chart: the hub is the CEO, connectors are department heads, and interior nodes are employees. Each group has uniform connectivity within its role, so the whole matrix collapses to a 3×3 summary.

Relation to previous sections: Definition 3.4 introduces Caterpillar graphs as another family (alongside SoC) whose spectral properties will be compared in later results. Proposition 4.1 delivers the first *closed-form* spectral radius, directly using the SoC construction from Section 3 and the equitable partition technique implicit in prior structural definitions.

The equitable partition reduces a potentially large Laplacian to a 3×3 quotient matrix, making an exact closed-form spectral radius tractable regardless of graph size.

ORIGINAL (ENGLISH)

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^3}{d_v^2}, \tag{5}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 4$ *and* $t \geq 2m - 1$. *In particular,* $\mathrm{SoC}(K_8, 19)$ *with* $n = 153$ *satisfies* $\mu(G) \approx 20.557$ *while the bound gives* ≈ 19.191.

Proof. By Proposition 4.1, we have $\mu = \frac{(m+t+1)+\sqrt{(t-m)^2+2(t+m)-3}}{2}$. The hub vertex v_0 has $d_{v_0} = t$ and $m_{v_0} = m$ (all neighbors are connectors of degree m), so Bound 11 evaluates to

$$\frac{2m_{v_0}^3}{d_{v_0}^2} = \frac{2m^3}{t^2}.$$

EXPLANATION (ENGLISH)

Theorem 4.2 — Violating BHS Bound 11

Variable/Symbol Explanation

- $\mu(G)$: largest Laplacian eigenvalue of graph G
- m_v: average degree of neighbors of vertex v
- d_v: degree of vertex v
- v_0: **hub vertex** of $\mathrm{SoC}(K_m, t)$, with $d_{v_0} = t, m_{v_0} = m$

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^3}{d_v^2} \quad \text{(Bound 11)}$$

$$\mu = \frac{(m+t+1)+\sqrt{(t-m)^2+2(t+m)-3}}{2}$$

At v_0, Bound 11 collapses to $\frac{2m^3}{t^2}$. For $m = 8,\ t = 19$: bound $\approx 19.191 < \mu \approx 20.557$. Counterexample confirmed.

Think of Bound 11 as a speed limit estimated from local traffic near each intersection. The hub vertex v_0 is a highway on-ramp where the local estimate is too optimistic — actual throughput (eigenvalue) exceeds the posted limit.

Connection to previous sections: Proposition 4.1 (equation 3) derived the closed-form μ via the piecewise-constant eigenvector on cells — that formula is directly substituted here to produce the violation. Reference [4] established this same refutation task; Theorem 4.2 is its concrete payoff.

The hub vertex, by concentrating all cross-edges, is the single witness that breaks the bound — one vertex suffices to invalidate a global upper-bound conjecture.

v_0

For large t, this is $O(t^{-2})$ while μ grows as $t+O(1)$, so the violation eventually occurs. Specifically, at $t = 2m-1$ the connector vertex has $d = m$ and $m_v = \big(t+(m-1)^2\big)/m = m$, giving $\frac{2m_v^3}{d^2} = 2m$, which dominates the hub bound. By Proposition 4.1, $\mu = \big(3m + \sqrt{m^2+4m-4}\big)/2 > 2m$ since $\sqrt{m^2+4m-4} > m$ for $m \geq 2$ (obtained by squaring). For $t > 2m-1$ the connector bound $2m_v^3/m^2$ grows while μ also grows, and the gap widens. □

Theorem 4.3. *BHS Bound 13,*

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3}, \tag{6}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and* $t = 2m-1$. *In particular,* $\mathrm{SoC}(K_9, 19)$ *with* $n = 172$ *satisfies* $\mu(G) \approx 20.685$ *while the bound gives* ≈ 19.845.

Proof. The connector vertex has $d = m$ and $m_v = \big(t + (m-1)^2\big)/m$. At $t = 2m-1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\mathrm{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m + \sqrt{m^2+4m-4})/2$. Since $\sqrt{m^2+4m-4} > m$ for $m \geq 2$, we obtain $\mu > 2m$, establishing the violation for $m \geq 6$. For $m = 5$

Theorem 4.3 — BHS Bound Violation by SoC(K_m, t)

This section proves that **BHS Bound 13** — another published spectral upper bound — is also violated by the SoC(K_m, t) family, extending the counterexample campaign from Propositions 4.1–4.2.

Equations & Variables

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3} \tag{6}$$

- $\mu(G)$: largest adjacency eigenvalue (spectral radius)
- d_v: degree of vertex v
- m_v: number of edges incident to neighbors of v (local edge count)
- m, t: construction parameters of SoC(K_m, t)

Key computation. For the connector vertex at $t = 2m-1$: $d = m$ and $m_v = \lfloor (t + (m-1)^2)/m \rfloor = m$, so the bound gives $2m^8/m^7 = 2m$. Meanwhile by Proposition 4.1:

$$\mu = \frac{3m + \sqrt{m^2+4m-4}}{2} > 2m \quad (m \geq 2)$$

since $\sqrt{m^2+4m-4} > m$ (verified by squaring). The concrete witness is SoC(K_9, 19) with $n = 172$: $\mu \approx 20.685$ vs. bound ≈ 19.845.

ORIGINAL (ENGLISH)

v_0

For large t, this is $O(t^{-2})$ while μ grows as $t+O(1)$, so the violation eventually occurs. Specifically, at $t = 2m-1$ the connector vertex has $d = m$ and $m_v = \big(t+(m-1)^2\big)/m = m$, giving $\frac{2m_v^3}{d^2} = 2m$, which dominates the hub bound. By Proposition 4.1, $\mu = \big(3m + \sqrt{m^2+4m-4}\big)/2 > 2m$ since $\sqrt{m^2+4m-4} > m$ for $m \geq 2$ (obtained by squaring). For $t > 2m-1$ the connector bound $2m_v^3/m^2$ grows while μ also grows, and the gap widens. □

Theorem 4.3. *BHS Bound 13,*

$$\mu(G) \leq \max_{v \in V} \frac{2m_v^4}{d_v^3}, \tag{6}$$

is violated by $\mathrm{SoC}(K_m, t)$ *for all* $m \geq 5$ *and* $t = 2m-1$. *In particular,* $\mathrm{SoC}(K_9, 19)$ *with* $n = 172$ *satisfies* $\mu(G) \approx 20.685$ *while the bound gives* ≈ 19.845.

Proof. The connector vertex has $d = m$ and $m_v = \big(t + (m-1)^2\big)/m$. At $t = 2m-1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\mathrm{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m + \sqrt{m^2+4m-4})/2$. Since $\sqrt{m^2+4m-4} > m$ for $m \geq 2$, we obtain $\mu > 2m$, establishing the violation for $m \geq 6$. For $m = 5$

EXPLANATION (ENGLISH)

Think of the bound as a speed camera calibrated for normal traffic. SoC's connector vertex is a vehicle that looks ordinary by local measurements (degree m, edge count m) yet moves faster than the camera expects — the bound underestimates the global eigenvalue.

Relation to previous sections. Proposition 4.1 supplied the closed-form μ; Proposition 4.2 violated an earlier hub-based bound using the same graph family. Theorem 4.3 targets a *different* bound (degree-cubed denominator instead of degree-squared), showing the SoC construction defeats multiple independent spectral inequalities from the literature.

One graph family, one carefully chosen parameter $t = 2m-1$, is enough to simultaneously falsify multiple published upper bounds on μ.

$t = 2m - 1$ we have $t + (m-1)^2 = m^2$, giving $B_{13}^{\text{conn}} = 2m^8/m^7 = 2m$. By Proposition 4.1, $\mu = (3m + \sqrt{m^2 + 4m - 4})/2$. Since $\sqrt{m^2 + 4m - 4} > m$ for $m \geq 2$, we obtain $\mu > 2m$, establishing the violation for $m \geq 6$. For $m = 5$, the interior bound is ≈ 10.195 but $\mu(\mathrm{SoC}(K_5, 9)) \approx 10.702 > 10.195$. □

Proposition 4.4. *For $k \geq 2$, the Laplacian spectral radius of* $\mathrm{DS}(k,k)$ *is*

$$\mu(\mathrm{DS}(k,k)) = \frac{(k+3) + \sqrt{k^2 + 6k + 1}}{2}. \tag{7}$$

Proposition 4.4 — Laplacian Spectral Radius of DS(k, k)

This section closes the verification started earlier (m ≥ 5 cases) and derives a **closed-form spectral radius** for the double-star graph **DS(k, k)** via symmetry decomposition.

Variable/Symbol Explanation

$$\mu(DS(k,k)) = \frac{(k+3) + \sqrt{k^2 + 6k + 1}}{2} \tag{7}$$

- μ: largest eigenvalue of the Laplacian matrix (spectral radius)
- $DS(k,k)$: double-star graph — two hubs each connected to k pendant leaves
- Q^+, Q^-: 2×2 quotient matrices for the symmetric/antisymmetric eigenspaces
- λ^+: larger root of the antisymmetric characteristic polynomial

Proof Structure

The **Z_2 symmetry** (swapping the two hubs) splits the eigenspace into two sectors:

- **Symmetric** Q^+: eigenvalues 0 and $k+1$
- **Antisymmetric** Q^-: characteristic polynomial $\lambda^2 - (k+3)\lambda + 2 = 0$, giving $\lambda^+ = \frac{(k+3)+\sqrt{k^2+6k+1}}{2}$
- Pendant leaves add eigenvalue 1 with multiplicity $2(k-1)$, dominated by λ^+

ORIGINAL (ENGLISH)

Proof. The $\mathbb{Z}_2$ symmetry swapping the two hubs decomposes eigenspaces into symmetric and antisymmetric sectors.

Symmetric sector. Quotient matrix $Q_+ = \left(\begin{smallmatrix} k & -k \\ -1 & 1 \end{smallmatrix}\right)$, eigenvalues 0 and $k+1$.

Antisymmetric sector. Quotient matrix $Q_- = \left(\begin{smallmatrix} k+2 & -k \\ -1 & 1 \end{smallmatrix}\right)$, characteristic polynomial $\lambda^2 - (k+3)\lambda + 2 = 0$, eigenvalues $(k+3 \pm \sqrt{k^2+6k+1})/2$.

Pendant leaves contribute eigenvalue 1 with multiplicity $2(k-1)$. Since $\sqrt{k^2+6k+1} > k-1$ for $k \geq 2$, the antisymmetric λ_+ is the global maximum. □

EXPLANATION (ENGLISH)

Think of the two-hub graph as a seesaw. The Z_2 symmetry lets you analyze "both sides move together" (symmetric) and "sides move opposite" (antisymmetric) independently — the antisymmetric mode is always stiffer, producing the larger eigenvalue.

Connection to Previous Sections

Earlier sections established an interior bound on μ and verified violations for SoC graphs (m $\geq$ 5, 6). Proposition 4.4 provides the exact μ for DS(k,k) as a **benchmark comparison target** — the same spectral radius framework, now solved in closed form rather than numerically approximated.

Symmetry decomposition turns an n×n eigenvalue problem into two 2×2 problems, making the exact closed form (eq. 7) tractable.

$$\mu(G) \le \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right), \quad (8)$$

is violated by $\mathrm{DS}(k,k)$ *for all* $k \ge 6$. *At* $k = 6$, $\mathrm{DS}(6,6)$ *with* $n = 14$ *satisfies* $\mu(G) \approx 8.772$ *while the bound gives* ≈ 8.676.

Proof. The hub–leaf edge maximizes Bound 45 for $k \ge 6$. Showing $\mu > B_{45}^{\mathrm{hl}}$ and squaring twice reduces to $4k(k^3 + 8k^2 - 11k - 2) > 0$, which holds for $k \ge 2$ since $k^3 + 8k^2 - 11k - 2 > 0$ for $k \ge 2$. The gap satisfies $B_{45} - \mu = -1/(2k) + O(k^{-2})$. □

Proposition 4.6. *For* $t \ge 2$ *and* $s \ge 1$, *the Laplacian spectral radius of* $\mathrm{SoS}(t,s)$ *is*

$$\mu(\mathrm{SoS}(t,s)) = \frac{(t+s+2) + \sqrt{(t-s)^2 + 4s}}{2}. \quad (9)$$

Proof. Equitable partition into {hub}, {intermediates}, {leaves}. The quotient Laplacian is $Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & 1+s & -s \\ 0 & -1 & 1 \end{pmatrix}$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

Theorem 4.5 & Proposition 4.6 — Violations and Exact Spectral Radius

Variable/Symbol Explanation for the bound (8) and (9):

- $\mu(G)$: **Laplacian spectral radius** — largest eigenvalue of the Laplacian matrix
- d_i, d_j: degrees of endpoint vertices i, j of edge (i, j)
- m_i, m_j: average neighbor degrees of i and j
- $DS(k, k)$: double-star graph — two hubs each with k leaves, joined by one edge
- $SoS(t, s)$: star-of-stars — hub connected to t intermediates, each with s leaves

$$\mu(G) \le \max_{(i,j)\in E} \frac{2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}}{2} \quad (8)$$

$$\mu(SoS(t,s)) = \frac{(t+s+2) + \sqrt{(t-s)^2 + 4s}}{2} \quad (9)$$

Concept Relationships: Earlier sections introduced spectral bounds as candidate upper envelopes for $\mu(G)$. Theorem 4.5 now shows Bound 45 is not universally tight — $DS(k, k)$ escapes it for $k \ge 6$, motivating the search for sharper bounds. Proposition 4.6 provides the *exact* value for $SoS(t, s)$ via an equitable partition, giving a clean benchmark to test future bounds against.

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$$\mu(G) \le \max_{(i,j)\in E} \left(2 + \sqrt{(d_i - d_j)^2 + 2(d_i m_i + d_j m_j) - 4(m_i + m_j) + 4}\right), \quad (8)$$

is violated by $\mathrm{DS}(k,k)$ *for all* $k \ge 6$. *At* $k = 6$, $\mathrm{DS}(6,6)$ *with* $n = 14$ *satisfies* $\mu(G) \approx 8.772$ *while the bound gives* ≈ 8.676.

Proof. The hub–leaf edge maximizes Bound 45 for $k \ge 6$. Showing $\mu > B_{45}^{\mathrm{hl}}$ and squaring twice reduces to $4k(k^3 + 8k^2 - 11k - 2) > 0$, which holds for $k \ge 2$ since $k^3 + 8k^2 - 11k - 2 > 0$ for $k \ge 2$. The gap satisfies $B_{45} - \mu = -1/(2k) + O(k^{-2})$. □

Proposition 4.6. *For* $t \ge 2$ *and* $s \ge 1$, *the Laplacian spectral radius of* $\mathrm{SoS}(t,s)$ *is*

$$\mu\big(\mathrm{SoS}(t,s)\big) = \frac{(t+s+2) + \sqrt{(t-s)^2 + 4s}}{2}. \quad (9)$$

Proof. Equitable partition into {hub}, {intermediates}, {leaves}. The quotient Laplacian is $Q_L = \begin{pmatrix} t & -t & 0 \\ -1 & 1+s & -s \\ 0 & -1 & 1 \end{pmatrix}$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

EXPLANATION (ENGLISH)

Think of Bound 45 as a speed limit sign: most drivers (graphs) stay under it, but $DS(k,k)$ is a car that legally exceeds the posted limit — exposing that the sign was placed incorrectly. The proof reduces the violation to a polynomial inequality $k^3 + 8k^2 - 11k - 2 > 0$, a simple algebraic certificate.

The gap $B_{45} - \mu = -\frac{1}{2k} + O(k^{-2})$ quantifies how badly the bound fails: it grows tighter as $k \to \infty$ but remains negative for all finite $k \ge 6$.

Exact closed-form spectral radii like eq. (9) are rare and valuable — they serve as hard test cases that any new bound must survive.

tient Laplacian is $Q_L = \begin{pmatrix} -1 & 1+s & -s \\ 0 & -1 & 1 \end{pmatrix}$ with characteristic polynomial $-\lambda[\lambda^2 - (t+s+2)\lambda + n]$, where $n = 1 + t(1+s)$. Discriminant $(t-s)^2 + 4s > 0$. □

Theorem 4.7. *BHS Bound 40,*

$$\mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right), \tag{10}$$

is violated by SoS(39, 8) *with* $n = 352$, *where* $\mu(G) \approx 40.256$ *and the bound gives* ≈ 38.982.

Proposition 4.6 Tail + Theorem 4.7 & Remark 4.8 — BHS Bounds Violated

Connection to previous section: The opening line closes Proposition 4.6's characteristic polynomial. The discriminant $(t-s)^2 + 4s > 0$ guarantees two distinct real eigenvalues — the exact formula Theorem 4.7 then plugs into a counterexample.

- $\mu(G)$: largest Laplacian eigenvalue of G
- d_i, d_j: degrees of the two endpoints of edge (i, j)
- m_i, m_j: average neighbor degree at vertex i, j
- B_{40}, B_{48}: the numeric upper-bound values from BHS Bounds 40 and 48

$$\text{Bound 40: } \mu(G) \leq \max_{(i,j)\in E} \left(2 + \sqrt{2(m_i-1)^2 + 2(m_j-1)^2 + d_i^2 + d_j^2 - d_i m_i - d_j m_j}\right)$$

$$\text{Bound 48: } \mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}}$$

Proof sketch (Thm 4.7): For the hub–intermediate edge of **SoS(39,8)** ($n = 352$): $d_h = 39,\ m_h = 9,\ d_w = 9,\ m_w = \frac{47}{9}$. Proposition 4.6 yields

$$\mu = \frac{49 + \sqrt{993}}{2} \approx 40.256, \quad B_{40} \approx 38.982$$

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Proof. The hub–intermediate edge has $d_h = 39$, $m_h = 9$, $d_w = 9$, $m_w = 47/9$. By Proposition 4.6, $\mu = (49 + \sqrt{993})/2$. The inequality $\mu > B_{40}$ reduces to $\sqrt{993} > 27.25$, i.e., $993 > 742.6$, which holds. □

Remark 4.8 (Bound 48 — degenerate case). BHS Bound 48,

$$\mu(G) \leq \max_{(i,j)\in E} \frac{2(d_i^2 + d_j^2)}{2 + \sqrt{2(d_i^2 + d_j^2) - 4(m_i + m_j) + 4}},$$

is violated by P_3: the bound gives $10/(2+\sqrt{2}) \approx 2.929$ while $\mu(P_3) = 3$. This is the only counterexample found.

5. Why these bounds fail

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Violation confirmed since $\sqrt{993} > 27.25$, i.e., $993 > 742.6$.

Remark 4.8: P_3 (path on 3 vertices, $\mu(P_3) = 3$) gives $B_{48} = \frac{10}{2+\sqrt{2}} \approx 2.929 < 3$. This is the only known counterexample to Bound 48, flagged as a degenerate edge case.

Each BHS bound is like a bridge load rating computed from local road stats. Both ratings are exceeded here — not catastrophically, but enough to prove the formula underestimates reality on certain graph structures.

Two distinct BHS spectral bounds fail on explicit graphs, motivating Section 5's structural explanation of why local degree statistics cannot universally cap global Laplacian eigenvalues.

5.1. Hub-invisible bounds

Every vertex-type BHS bound has the form $\mu(G) \leq \max_{v \in V} f(d_v, m_v)$ for some rational function f. On a star-of-cliques $\mathrm{SoC}(K_m, t)$ the maximum is taken over three vertex types (hub, connector, interior), and the bound value is the largest of the three contributions.

Definition 5.1 (Hub-invisible). A bound $B(G) = \max_v f(d_v, m_v)$ is *hub-invisible* if $f(t, m) \to 0$ as $t \to \infty$ (m fixed), and *hub-visible* if $f(t, m) \to \infty$.

Proposition 5.2 (Hub-invisible bounds fail). (a) ***Bounds 11, 13*** *are hub-invisible:* $f_{11} = 2m^3/t^2 \to 0$, $f_{13} = 2m^4/t^3 \to 0$.

(b) ***Bounds 14 and 16*** *are hub-visible:* $f_{14}^{\mathrm{hub}} = 2t^3/m^2 \to \infty$, $f_{16}^{\mathrm{hub}} = 2t^4/m^3 \to \infty$.

Section 5.1 — Hub-invisible Bounds

Variable/Symbol Explanation

- $\mu(G)$: spectral bound being estimated (e.g., largest eigenvalue)
- $f(d_v, m_v)$: rational function of vertex degree d_v and local parameter m_v
- t: hub degree in **SoC**(K_m, t) — grows large in the limit
- m: clique size (fixed while $t \to \infty$)

> $$\mu(G) \leq \max_{v \in V} f(d_v, m_v)$$

Concept Relationships

Previous sections introduced specific numbered BHS-type bounds (11, 13, 14, 16, 30) and the **star-of-cliques** SoC(K_m, t) as a test graph. Here, those bounds are *classified* by their sensitivity to the hub vertex — a structural diagnostic, not a new bound.

> Think of the hub as a celebrity node with t connections. A hub-invisible bound "ignores" the celebrity: as their follower count explodes, the bound's contribution from that node shrinks to zero — making the bound blind to the most influential vertex. A hub-visible bound correctly registers the celebrity's outsized role.

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5.1. Hub-invisible bounds

Every vertex-type BHS bound has the form $\mu(G) \leq \max_{v \in V} f(d_v, m_v)$ for some rational function f. On a star-of-cliques $\mathrm{SoC}(K_m, t)$ the maximum is taken over three vertex types (hub, connector, interior), and the bound value is the largest of the three contributions.

Definition 5.1 (Hub-invisible). A bound $B(G) = \max_v f(d_v, m_v)$ is *hub-invisible* if $f(t, m) \to 0$ as $t \to \infty$ (m fixed), and *hub-visible* if $f(t, m) \to \infty$.

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(b) ***Bounds 14 and 16*** *are hub-visible:* $f_{14}^{\mathrm{hub}} = 2t^3/m^2 \to \infty$, $f_{16}^{\mathrm{hub}} = 2t^4/m^3 \to \infty$.

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Definition 5.1 — Hub-invisible vs. Hub-visible

A bound $B(G) = \max_v f(d_v, m_v)$ is **hub-invisible** if $f(t, m) \to 0$ as $t \to \infty$ (m fixed); **hub-visible** if $f(t, m) \to \infty$.

Proposition 5.2 summary:

- Bounds 11, 13: hub-invisible — $f_{11} = 2m^3/t^2,\ f_{13} = 2m^4/t^3 \to 0$
- Bounds 14, 16: hub-visible — $\sim t^3/m^2,\ t^4/m^3 \to \infty$
- Bound 30: hub-visible — dominant term $t^2/m \to \infty$

Hub-invisibility is a *structural failure mode*: any bound that vanishes at the hub vertex is provably loose on star-of-cliques graphs, regardless of how tight it looks on other families.

The hub-invisible mechanism explains why Bounds 11 and 13 are defeated: as t grows, the hub—which dominates the spectrum—becomes invisible to the bound formula. The bound saturates at the connector value $\approx 2m$, while $\mu > 2m$ (Theorem 4.2).

Remark 5.3. Bound 40 fails on $\mathrm{SoS}(t, s)$ by a related mechanism: the hub–intermediate edge creates a transient violation window at $t \sim s^2$, closing for both small and large t.

5.2. Edge-based antisymmetric modes

Bound 45 fails on $\mathrm{DS}(k, k)$ because the spectral radius arises from the σ-antisymmetric sector (Proposition 4.4), where the two hubs carry opposite signs. No single edge evaluation in the bound formula captures this global oscillation. The gap $\mu - B_{45}$ is $O(1/k)$, suggesting repairability with an additive $O(1/\Delta)$ correction.

§ 5.1–5.2 — Why Specific Bounds Fail: Hub-Invisibility & Antisymmetric Modes

Concept Relationships: § 4 established the **hub-and-connector** spectral structure and proved $\mu > 2m$ (Theorem 4.2). Here, § 5 diagnoses precisely *why* known bounds cannot see that gap — two failure modes emerge.

Failure Mode 1: Hub-Invisibility (Bounds 11, 13, 40)

As t grows, the hub dominates μ but becomes **invisible** to the bound formula, which saturates at the connector value $\approx 2m$. Since $\mu > 2m$, the bound is perpetually loose.

- μ: true spectral radius of the graph
- $2m$: saturation value of the bound (connector-level estimate)
- t: growth parameter; as $t \to \infty$, hub dominance increases
- $t \sim s^2$: transient violation window in $\mathrm{SoS}(t, s)$ where Bound 40 also fails

> Think of the bound formula as a speedometer that only reads local road speed limits — as soon as a highway (the hub) enters the picture, the speedometer saturates at the last posted sign and never registers the true velocity.

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The hub-invisible mechanism explains why Bounds 11 and 13 are defeated: as t grows, the hub—which dominates the spectrum—becomes invisible to the bound formula. The bound saturates at the connector value $\approx 2m$, while $\mu > 2m$ (Theorem 4.2).

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Failure Mode 2: Antisymmetric Oscillation (Bound 45 on $\mathrm{DS}(k, k)$)

In the **σ-antisymmetric sector**, the two hubs carry *opposite signs*. No single-edge evaluation in Bound 45 can capture this global oscillation — it requires seeing both hubs simultaneously.

- $\mathrm{DS}(k, k)$: double-star graph, two hubs connected by a bridge
- σ-antisymmetric: eigenvector sector where hub amplitudes differ in sign
- $\mu - B_{45} = O(1/k)$: gap shrinks with k, hinting the bound is nearly repairable

Bound 45 is like judging a seesaw's imbalance by measuring only one side — without the opposing hub's sign, the antisymmetric "lift" is invisible.

$$\mu - B_{45} = O\left(\frac{1}{k}\right) \implies B_{45} + O\left(\frac{1}{\Delta}\right) \text{ could close the gap}$$

The two failure modes reveal a structural divide: hub-invisible bounds miss spectral mass that accumulates at a single dominant node, while edge-local bounds miss eigenvector *sign patterns* that span multiple hubs — both are fixable but require fundamentally different corrections.

5.3. Characterizing vulnerable bounds

Conjecture 5.4. *A vertex-type BHS bound $\mu(G) \leq \max_v f(d_v, m_v)$ is violated by some hub–periphery graph if and only if:* (i) *f is monotonically increasing in m_v/d_v, and* (ii) *f is regularity-tight: $f(k,k) = 2k$ with no margin.*

6. Classification of the remaining 33 bounds

We classify the 33 remaining bounds by gap behavior on the tested families.

The SUPER-SAFE bounds include all hub-visible bounds (14, 16, 30). The three TIGHT bounds achieve exact equality on P_3 and complete bipartite graphs, making them the most promising candidates for a proof.

7. Discussion

Summary.. BHS [2] proposed 68 conjectured bounds for $\mu(G)$. Ghebleh et al. [4] disproved 30 via cross-entropy methods. We have disproved 5 more by structural construction, leaving 33 open.

§ 5.3 & § 6–7: Vulnerable Bounds, Classification, and Summary

Variable/Symbol Explanation for Conjecture 5.4:

> $$\mu(G) \leq \max_v f(d_v, m_v)$$

- $\mu(G)$: spectral radius (largest eigenvalue) of graph G
- d_v: degree of vertex v
- m_v: average degree of v's neighbors
- $f(d_v, m_v)$: bound formula evaluated per vertex, maximized over all v
- $f(k,k) = 2k$: **regularity-tight** condition — no slack on regular graphs

The conjecture identifies exactly *which* vertex-type bounds are vulnerable to the hub-periphery counterexamples built in § 5.1–5.2. Two simultaneous conditions must hold: monotone increase in m_v/d_v (so hubs with high-degree neighbors are penalized more) **and** zero margin on regular graphs.

> Think of f as a speed limit that tightens on highways (high m_v/d_v) but is already at its legal minimum in quiet suburbs (regular graphs). A bound with both properties has no buffer — a hub-periphery graph exploits both loopholes at once.

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5.3. Characterizing vulnerable bounds

Conjecture 5.4. *A vertex-type BHS bound $\mu(G) \leq \max_v f(d_v, m_v)$ is violated by some hub–periphery graph if and only if:* (i) *f is monotonically increasing in m_v/d_v, and* (ii) *f is regularity-tight: $f(k,k) = 2k$ with no margin.*

6. Classification of the remaining 33 bounds

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7. Discussion

Summary.. BHS [2] proposed 68 conjectured bounds for $\mu(G)$. Ghebleh et al. [4] disproved 30 via cross-entropy methods. We have disproved 5 more by structural construction, leaving 33 open.

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Connection to previous sections: § 5.1 constructed the hub-periphery family; § 5.2 showed specific bounds fail on it. Conjecture 5.4 now generalizes this into a *characterization* — a two-condition litmus test predicting which of the remaining 33 bounds will fall to the same construction.

§ 6 partitions the 33 surviving bounds into two classes:

- **Super-safe**: hub-visible bounds (indices 14, 16, 30) — structurally immune
- **Tight**: achieve equality on P_3 and complete bipartite graphs — prime proof targets

> The tight bounds are the most mathematically valuable survivors: they cannot be improved, so a proof of any one would be sharp.

§ 7 Summary closes the arc: BHS proposed 68 bounds → Ghebleh et al. disproved 30 (cross-entropy) → this paper disproves 5 more (structural construction) → **33 remain open**.

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Structural vs. computational approaches.. Cross-entropy search explores adjacency matrices for $n \leq 24$. Our smallest counterexample has $n = 29$, beyond exhaustive reach. Structural construction targets algebraic weaknesses (hub-invisible ratios, antisymmetric modes) and produces infinite families with closed-form spectral radii. The two approaches are complementary.

Gap asymptotics.. Bounds 11/13: gap diverges (irreparable). Bound 40: transient window (max gap 1.74). Bound 45: gap $\to 0$ as $1/(2k)$ (repairable). Bound 48: degenerate P_3 case only.

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Structural vs. Computational: Method Comparison & Gap Asymptotics

Concept Relationships: Earlier sections introduced the two attack strategies independently — cross-entropy search (computational) and algebraic construction (structural). This passage draws the dividing line: **n = 29** is the boundary where exhaustive search fails and closed-form construction takes over.

- **n ≤ 24**: adjacency matrix space tractable by cross-entropy search
- **n = 29**: smallest confirmed counterexample — unreachable by brute force
- **hub-invisible ratios / antisymmetric modes**: algebraic targets exploited by structural construction to yield infinite families

> Think of it like lock-picking vs. key-forging: cross-entropy search tries every key up to size 24; structural construction understands the lock's mechanism and forges keys of any size, including 29+.

Gap Asymptotics Summary:

- **Bounds 11/13**: gap diverges — the bound is permanently broken
- **Bound 40**: transient failure, max gap ≈ 1.74 then recovers
- **Bound 45**: gap $\to 0$ as $\frac{1}{2k}$ — asymptotically repairable
- **Bound 48**: failure limited to degenerate P_3 case only

> The asymptotic classification reveals that not all bound violations are equal — some are structural flaws, others are finite anomalies, directly informing which bounds are worth salvaging.

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Open questions.. (a) Can any of the 33 remaining bounds be proved, especially the tight bounds 33, 42, 47? (b) Do three-level hierarchies violate additional bounds? (c) Can Conjecture 5.4 be proved? (d) Can Bound 45 be repaired with an $O(1/\Delta)$ correction?

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the author used Claude (Anthropic; model: Claude Opus 4, 2025) for computational assistance, including code development for graph construction and eigenvalue computation, structural analysis of graph families, and systematic counterexample search. The author reviewed and edited all content and takes full responsibility for the publication.

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Open Questions & AI Declaration

This closing section poses **four open problems** distilled from the paper's unresolved bounds, then formally discloses AI-assisted methodology.

- **(a)** Proving any of the 33 remaining bounds — especially the suspected tight bounds 33, 42, 47
- **(b)** Testing whether **three-level hierarchies** violate further spectral bounds beyond those already found
- **(c)** Establishing **Conjecture 5.4** rigorously
- **(d)** Repairing **Bound 45** via an $O(1/\Delta)$ correction term

> Think of the 33 unproved bounds as a mountain range where some peaks (tight bounds 33, 42, 47) are already mapped but unclimbed — question (a) asks who can summit first, while (d) asks whether a known broken trail can be rerouted by a small detour of order $1/\Delta$.

Relation to previous sections: The counterexample searches (Sections 4–5) eliminated candidate bounds and confirmed tightness of several others; these open questions are the direct residue — what counterexample search *could not* decide and what conjecture (5.4) remains formally unproven.

> The $O(1/\Delta)$ repair in question (d) suggests the bound's failure is quantifiably small — a structural clue that may guide the next proof attempt.

The **AI declaration** notes Claude Opus 4 assisted with graph construction code, eigenvalue computation, and counterexample search — the computational backbone of Sections 3–5 — while the author retains full intellectual and editorial responsibility.

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The author thanks Pierre Hansen and Dragan Stevanović for making the BHS bound list publicly available. Computational assistance was provided by Claude (Anthropic); all mathematical results were independently verified by the author.

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Acknowledgements & References

This closing section credits the human collaborators and computational tools that supported the research, then anchors every claim made throughout the paper to verifiable prior literature.

Acknowledgements

- **BHS bound list**: The precomputed database of Brouwer–Haemers–Stevanović spectral bounds used to validate conjectures in earlier sections was made public by Hansen and Stevanović.
- Claude (Anthropic) : Provided computational assistance; the author explicitly notes that all mathematical results were *independently verified*—a standard integrity disclaimer for AI-assisted research.

Reference Roles

- **[1] Brouwer & Haemers (2012)** — the foundational textbook on graph spectra; the theoretical backbone behind every eigenvalue bound discussed.
- **[2] Brankov, Hansen & Stevanović (2006)** — source of the automated conjecture framework and the BHS upper-bound list cited in the acknowledgements.
- **[3] Caporossi & Hansen (2000)** — introduces AutoGraphiX, the variable-neighborhood search system likely used to generate or cross-check extremal graph candidates.

> Think of the references as the paper's load-bearing walls: the new results built in the Method and Results sections only stand because they rest on these established structures. Remove any one of them and a key claim loses its foundation.

ORIGINAL (ENGLISH)

[1] A. E. Brouwer, W. H. Haemers, Spectra of Graphs, Universitext, Springer, 2012. doi:10.1007/978-1-4614-1939-6.

[2] V. Brankov, P. Hansen, D. Stevanović, Automated conjectures on upper bounds for the largest Laplacian eigenvalue of graphs, Linear Algebra and its Applications 414 (2–3) (2006) 407–424. doi:10.1016/j.laa.2005.10.017.

[3] G. Caporossi, P. Hansen, Variable neighborhood search for extremal graphs. 1. The AutoGraphiX system, Discrete Mathematics 212 (1–2) (2000) 29–44. doi:10.1016/S0012-365X(99)00206-X.

EXPLANATION (ENGLISH)

Relationship to Previous Sections: Every bound proved or conjectured in the main body traces back to these three works — [1] for spectral theory foundations, [2] for the BHS benchmark data used in comparisons, and [3] for the computational search strategy behind extremal examples.

> Explicitly separating "computational assistance" from "independent mathematical verification" signals that AI tools were used responsibly — accelerating search without replacing rigorous proof.

ORIGINAL (ENGLISH)

[4] M. Ghebleh, S. Al-Yakoob, A. Kanso, D. Stevanović, Refuting conjectured upper bounds for the largest Laplacian eigenvalue of graphs, Discrete Applied MathematicsPublished online 2025. Available at https://arxiv.org/abs/2403.18429 (2025).

[5] A. Z. Wagner, Constructions in combinatorics via neural networks, arXiv preprint arXiv:2104.14516 (2021).

[6] N. Bouffard, J. Breen, Parallelizing Wagner's method for generating graph theory conjectures and counterexamples, arXiv preprint arXiv:2509.01607 (2025).

[7] M. Roucairol, T. Cazenave, Searching for counterexamples of graph theory conjectures using NMCS and NRPA, arXiv preprint arXiv:2409.18626 (2024).

EXPLANATION (ENGLISH)

These four references form the **methodological lineage** of the paper's approach to automated conjecture refutation in graph theory.

- **[4]** Direct precedent work — refuting upper-bound conjectures on the **largest Laplacian eigenvalue**, the same target property as this paper.
- **[5]** Wagner (2021): original neural network search framework for combinatorial constructions — the algorithmic ancestor of the method used here.
- **[6]** Bouard & Breen (2025): parallelized extension of Wagner's method, improving scalability of counterexample generation.
- **[7]** Roucairol & Cazenave (2024): alternative search strategies (NMCS, NRPA) applied to the same class of problems.

> Think of these references as the "shoulders of giants" scaffold: [5] built the telescope, [6] added a wider lens, [7] tried a different mount, and [4] pointed them all at the same star this paper revisits.

Relationship to previous sections: Earlier sections introduced the paper's specific conjecture targets and experimental results. These citations ground those contributions — [4] establishes the prior domain result this work extends, while [5]–[7] situate the neural network search methodology within a growing research thread on AI-assisted discrete mathematics.

> The clustering of [5], [6], [7] around Wagner's framework signals that automated counterexample search via learned heuristics is now an active subfield, not an isolated trick.

ORIGINAL (ENGLISH)

[8] C. Godsil, G. Royle, Algebraic Graph Theory, Vol. 207 of Graduate Texts in Mathematics, Springer, 2001. doi:10.1007/978-1-4613-0163-9.

EXPLANATION (ENGLISH)

This is a **reference entry** for a foundational textbook on algebraic graph theory , supporting the spectral/graph-theoretic methods cited throughout the paper.

> Think of this as the dictionary the paper's graph math vocabulary comes from.

> Godsil & Royle underpins any eigenvalue or adjacency matrix arguments made in earlier sections.